

Phase-Adaptive Generation: Difficulty-Aware Semi-Autoregressive Decoding for Diffusion Language Models

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1 Problem and Motivation

Autoregressive (AR) models generate tokens one at a time, incurring high latency. Diffusion-based LLMs (dLLMs) such as Dream 7B (?) and LLaDA (?) generate tokens in parallel via iterative denoising, but apply uniform block sizes and denoising effort regardless of token difficulty. We observe that generation proceeds through **difficulty phases**: low-entropy stretches (e.g., “Ok, I will solve this”) alternate with high-entropy reasoning tokens (e.g., “derivative of x^2 is $2x$ ”). We propose **Phase-Adaptive Generation (PAG)**, a semi-AR decoding framework that detects these phases at inference time and adapts two dimensions per phase: (1) *chunk size* (tokens generated in parallel), and (2) *refinement intensity* (denoising steps per chunk). A lightweight learned MLP *proactively* predicts phase boundaries from hidden states, enabling anticipatory scheduling.

2 Related Work and Our Contribution

dLLMs. Dream 7B (?) matches AR baselines on general/math/code tasks and excels at planning (Countdown, Sudoku). Block Diffusion (?) introduced semi-AR decoding for dLLMs. **AdaBlock-dLLM** (?) (ICLR 2026) is the closest prior work: it adapts block size reactively using confidence-based “volatility band” detection, achieving up to 5.3% accuracy gains. However, it adjusts only *one* dimension (block size) and detects boundaries *after the fact*. **PAG** extends this along two axes: (i) a second adaptive dimension—refinement intensity—allocating more denoising compute to hard phases; (ii) proactive, learned phase prediction rather than reactive detection.

3 Experimental Plan

Models. Dream 7B (Base + Instruct) as primary dLLM; LLaDA-8B-Instruct as secondary. Qwen2.5-7B and LLaMA3-8B as AR baselines.

Benchmarks. GSM8K, MATH (math reasoning with clear easy/hard phase transitions); HumanEval, MBPP (code generation); Countdown, Sudoku (planning tasks where dLLMs excel). These align with AdaBlock-dLLM’s evaluation suite (?) for direct comparison.

Compute & Feasibility. We freeze the base dLLM entirely and only train a lightweight phase-prediction MLP ($\sim 100K$ parameters). Individual run estimates on a single A100 40GB: ~ 10 – 12 hrs to train the phase predictor per benchmark configuration, ~ 2 hrs for full evaluation per model–benchmark pair. Estimated total compute across all phases: ~ 120 GPU-hours.

Phased Plan. (1) Reproduce baselines + AdaBlock-dLLM (~ 20 hrs). (2) Analyze difficulty phases via hidden-state similarity and per-token entropy—this yields a standalone analytical contribution (~ 30 hrs). (3) Implement PAG: train phase predictor, adaptive chunk/refinement scheduling (~ 40 hrs). (4) Ablations: chunk-only vs. refinement-only, throughput–accuracy Pareto curves (~ 30 hrs).

Metrics. Task accuracy (exact-match, pass@1), throughput (tokens/sec), NFE, Pareto efficiency.

Risk Mitigation. If the learned predictor underperforms, we fall back to heuristic phase detection (confidence thresholds + clustering), still contributing the refinement-intensity dimension. Phase 2 alone provides a publishable characterization of difficulty phases in dLLM generation.

References

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